



Optimal Batting Lineup



TEAM MEMBERS

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COURSE & DATE

DASC 32003 · April 30, 2026

Why Does Lineup Order Matter?

Motivation

- $9! = 362,880$ possible orderings have vastly different run-scoring expectations.
- Batting order determines how often players reach high-leverage situations. Managers rely on tradition and intuition

Research Question

- Can an integer programming model identify a batting order that significantly outperforms a team's traditional lineup in expected runs per game — and what structural principles drive that improvement?

Scope & Assumptions

Scope:

- Optimizes Texas Rangers lineup using 2024 FanGraphs stats
- Builds 2 lineups: vs RHP and vs LHP
- Simultaneously assigns batting order **and** defensive positions
- Solved as a MILP using HiGHS via Pyomo

Assumptions:

- 2023-25 wOBA reflects true offensive ability going forward
- Platoon advantage calculated using wOBA vs LHP and vs RHP
- Plate appearance weights per lineup spot based on historical MLB averages
- Players only eligible at listed primary and secondary positions

Data Sources & Player Metrics

23

Roster
Players

3

Season(s)
of Data

29

Features
per Player

Data Sources & Citation

- ▶ FanGraphs.com — advanced sabermetric metrics

Key Sabermetric Parameters (c_{ij})

OBP — On-Base %

SLG — Slugging %

wRC+ — Weighted RC

wOBA — Weighted OBA

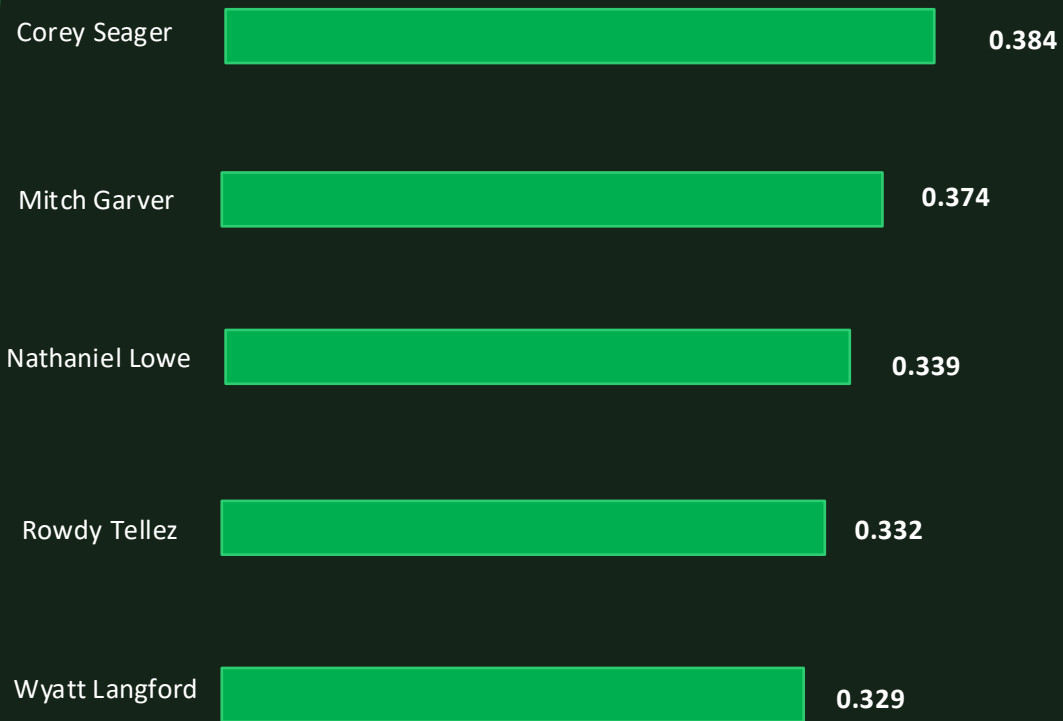
BB% — Walk %

L/R Platoon Split

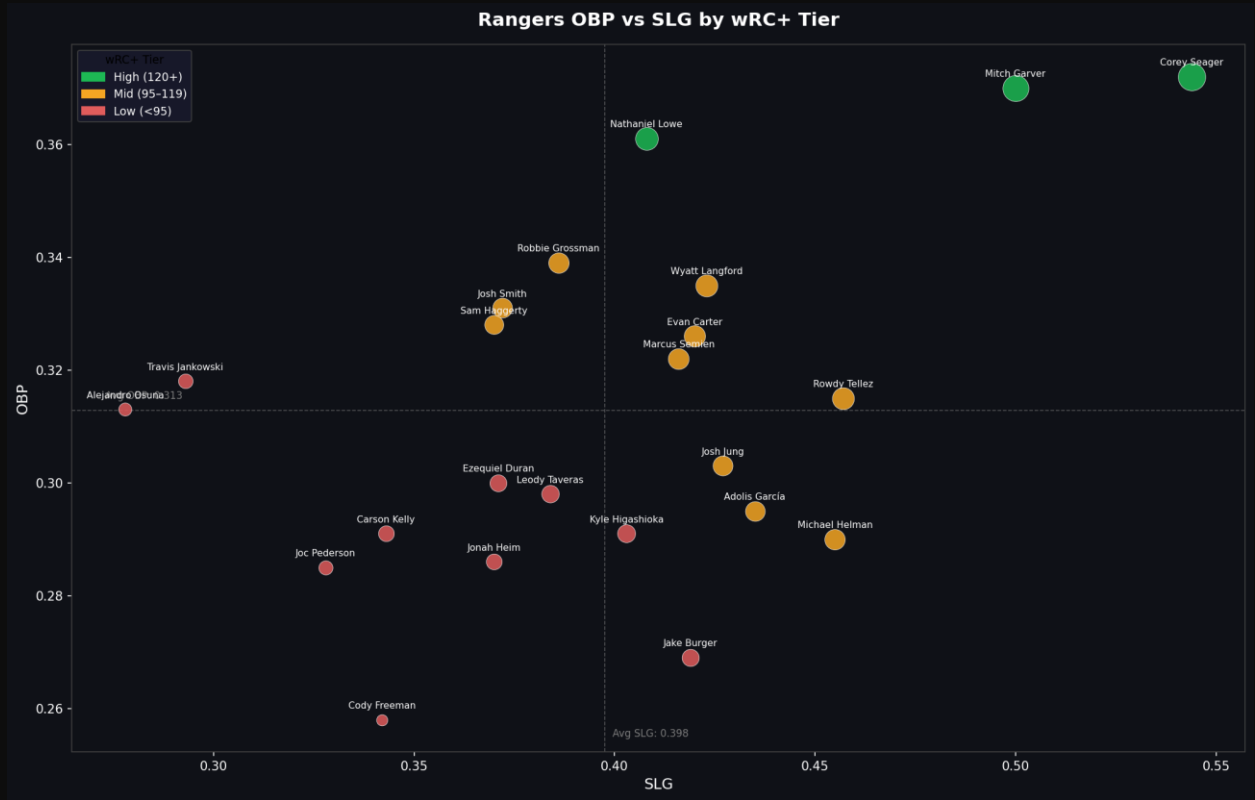
Defensive WAR per 162

Player Statistics at a Glance

wOBA by Player



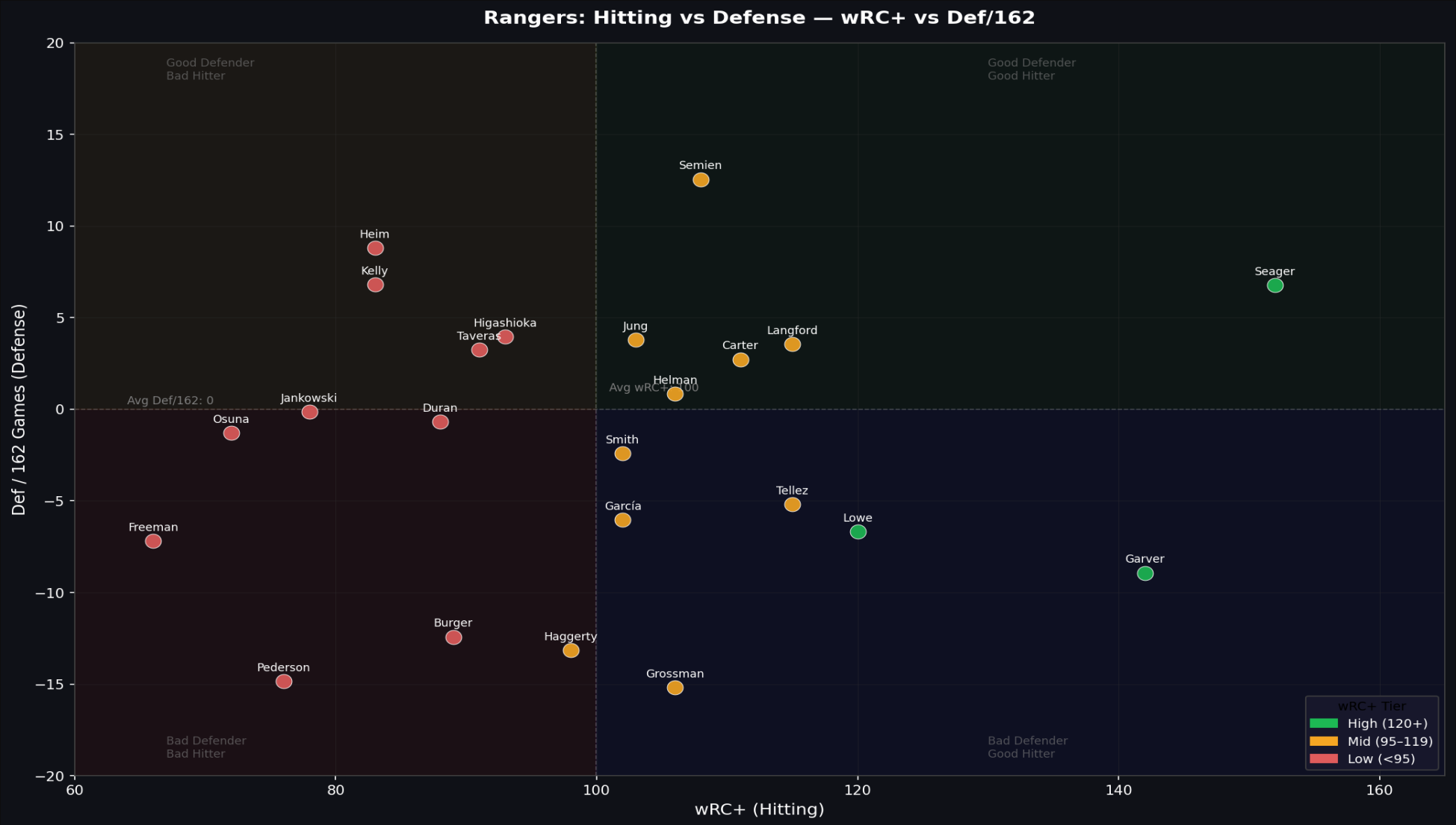
OBP vs. SLG by Player



High wRC+ Mid wRC+ Low wRC+

Player Statistics at a Glance - Extended

Defensive Ratings vs. Offensive Ratings



Decision Variables

Variable	Definition
$x_{i,s} \in \{0,1\}$	<i>Player i</i> bats in lineup spot <i>s</i>
$y_{i,f} \in \{0,1\}$	<i>Player i</i> plays fielding position <i>f</i>
$z_{i,j,s} \in \{0,1\}$	Synergy indicator: player <i>i</i> at spot <i>s</i> , player <i>j</i> follows at <i>s</i> + 1 or <i>s</i> + 2

Constraints

Description	Constraint
(1) One player per lineup spot:	$\forall s \in S: \sum_{i \in I} x_{i,s} = 1$
(2) Each player bats at most once:	$\forall i \in I: \sum_{s \in S} x_{i,s} \leq 1$
(3) One player per fielding position (DH optional):	$\sum_{i \in I} y_{i,DH} \leq 1 \quad \forall f \neq DH: \sum_{i \in I} y_{i,f} \leq 1$
(4) Each player at most one position:	$\forall i \in I: \sum_{f \in F} y_{i,f} \leq 1$
(5) Position Eligibility	$y_{i,f} \leq eligible_{i,f}$
(6) Linking Constraint – if a player is in the lineup, they must have exactly one position:	$\forall i: \sum_{s \in S} x_{i,s} = \sum_{f \in F} y_{i,f}$
(7) Synergy Activation – synergy fires only when player i and j are adjacent:	$z_{i,j,s} \leq x_{i,s}$ $z_{i,j,s} \leq \sum_{t \in next(s)} x_{j,t}$ $z_{i,j,s} \geq x_{i,s} \sum_{t \in next(s)} x_{j,t} - 1$

Objective Function & Components

Objective Function:

$$\text{Max } Z = \text{Offense} + \text{Synergy} + \text{Defense} - \text{Penalty}$$

(A): Offense – Platoon Adjusted wOBA:

$$\sum_{i \in I} \sum_{s \in S} w_s \cdot (q_{i,k} + wOBA_i)$$

(B): Synergy – Reward high-OBP hitter immediately before a power hitter:

$$\sum_{(i,j) \in \text{Pairs}} \sum_{s \in S} \alpha \cdot OBP_i \cdot Power_j \cdot z_{i,j,s}$$

(C): Defense – Reward good defenders at premium positions:

$$\sum_{i \in I} \sum_{f \in F} d \cdot def_i \cdot pos_f \cdot y_{i,f}$$

(D): Second Position Penalty – subtracted from objective:

$$\sum_{i \in I} \sum_{f \in F} \lambda \cdot y_{i,f} \cdot 1[f = \text{sec}(i)]$$

Sensitivity Analysis

Objective Coefficient Ranges

- The **smallest coefficient** (~ 0.0005) comes from synergy term: $\alpha \cdot OBP_i \cdot Power_j \cdot z_{i,j,s}$, where $\alpha = 0.05$ multiplied by the fractional OBP, giving us power scores that produced very small values
- The largest coefficient (~ 0.07) comes from the offensive term: $w_s \cdot q_{i,k} \cdot w_{i,s}$ - where the leadoff spot weight combined with a high wOBA player is responsible for the dominate term
- Given the range between the smallest and largest coefficients, offensive output drives most of the decisions the model makes, synergy and defense are meaningful, but rarely will override offensive ranking

RHS / Constraint Relaxation

Constraint	RHS	What relaxing it would do
(1) One player per spot	=1	Relaxing to ≤ 1 would allow empty lineup spots
(2) Each player bats at most once	≤ 1	Tightening closer to = would force all players to bat
(3) One per fielding position	= 1(non-DH), ≤ 1 (DH)	DH relaxed, for non-DH, relaxing to ≤ 1 would allow uncovered positions
(4) Each player at most one position	≤ 1	Relaxing would allow a player to play two position simultaneously
(5) Position Eligibility	$\leq$ eligible	Hard eligibility gate – relaxing would let players play positions that aren't specific to them.
(6) Linking Constraint	=	This critical structural constraint, relaxing would decipher batting order from defense only
(7) Synergy Activation	Three-constraint linearization	Relaxing lower bound would allow synergy credit only, and inflate objective

The Idea Behind the Model

Run Expectancy Matrix

- Each batting slot has a different expected plate appearances per game.
- Leadoff hitters bat $\sim 4.7x$ per game; the 9-hole bats $\sim 3.9x$. This slot weighting drives the coefficient $w(s)$ in the objective
- a wOBA point gained at spot 1 is worth more than the same gain at spot 9.
- Placing the best matchup hitters as high in the order as possible is the core intuition the model exploits.

Assignment Structure

- This is a variant of the **Binary Assignment Problem**. Each player is assigned to at most one batting spot and at most one fielding position per scenario.
- The model solves two linked assignment problems (batting order + defense) across two scenarios: vs RHP, vs LHP

Model Customization

- (1) **Platoon splits** — each hitter has a separate quality score $q_{i,k}$ for right- and left-handed pitchers, so the model naturally swaps players based on matchup.
- (2) **Defensive value** — the objective includes a small Def-based term, so the model avoids placing poor defenders at premium positions like SS, C, and CF.
- (3) **Synergy value** — the objective value rewards the model for having high on-base hitters before high power hitters to maximize runs created

Solving the Model using HiGHs

HiGHs Solver

- HiGHS is well-suited for large mixed-integer programs like this one, using branch-and-bound and cutting planes to find high-quality solutions quickly.
- Strong presolve and heuristics: Its presolve reductions and built-in heuristics significantly shrink the problem size and produce near-optimal solutions in seconds.
- The problem was split into 2 identical models for vs LHP and vs RHP. This allowed our model to run exponentially faster as it could complete one model fast then move on, rather than trying to solve multiple at once

Example Output

```
Solving report
Status          Optimal
Primal bound    0.45947193025
Dual bound      0.48216720298
Gap             4.94% (tolerance: 5%)
P-D integral    4.94776342384
Solution status feasible
                0.45947193025 (objective)
                0 (bound viol.)
                1.45994327738e-12 (int. viol.)
                0 (row viol.)

Timing          8.38
Max sub-MIP depth 3
Nodes           1
Repair LPs      0
LP iterations   10402
                0 (strong br.)
                3108 (separation)
                6440 (heuristics)
```

Termination vs RHP: optimal

```
Optimal lineup vs RHP:
Spot 1: Mitch Garver (DH)
Spot 2: Corey Seager (SS)
Spot 3: Rowdy Tellez (1B)
Spot 4: Adolis García (RF)
Spot 5: Marcus Semien (2B)
Spot 6: Josh Jung (3B)
Spot 7: Leody Taveras (CF)
Spot 8: Jonah Heim (C)
Spot 9: Wyatt Langford (LF)
Objective value: 0.45947193025042954
```

```
Solving report
Status          Optimal
Primal bound    0.462972259311
Dual bound      0.485530158148
Gap             4.87% (tolerance: 5%)
P-D integral    6.40194077893
Solution status feasible
                0.462972259311 (objective)
                0 (bound viol.)
                5.77315972805e-15 (int. viol.)
                0 (row viol.)

Timing          6.94
Max sub-MIP depth 4
Nodes           1
Repair LPs      0
LP iterations   8949
                0 (strong br.)
                2304 (separation)
                6086 (heuristics)
```

Termination vs LHP: optimal

```
Optimal lineup vs LHP:
Spot 1: Adolis García (RF)
Spot 2: Carson Kelly (C)
Spot 3: Joc Pederson (DH)
Spot 4: Josh Jung (3B)
Spot 5: Marcus Semien (2B)
Spot 6: Leody Taveras (CF)
Spot 7: Wyatt Langford (LF)
Spot 8: Corey Seager (SS)
Spot 9: Rowdy Tellez (1B)
Objective value: 0.46297225931079117
```

Optimal Lineup vs. Left Hand Pitcher

#	Player	Pos.	OBP	wRC+	Obj
1	Adolis Garcia	RF	.295	102	.0396
2	Carson Kelly	C	.291	83	.0517
3	Joc Pederson	DH	.285	76	.0424
4	Josh Jung	3B	.303	103	.0381
5	Marcus Semien	2B	.322	108	.0361
6	Leody Taveras	CF	.298	91	.0301
7	Wyatt Langford	LF	.335	115	.0361
8	Corey Seager	SS	.372	152	.0356
9	Rowdy Tellez	1B	.315	115	.0301

Obj. Value (IP)

Z = 0.4629

Solve Time: 8.1 seconds | LP Gap: 4.87%

Interpretation

Carson Kelly hits in the 2-hole vs Lefties for a couple reasons:

- .564 wOBA vs LHP as a Ranger in our data
- Solid defense at Catcher

Joc Pederson Bats 3rd against lefties despite being a LHB

- While playing for Texas, Pederson had reverse splits against lefties.

Corey Seager drops to 8:

- Hits lefties not as well as righties
- adds lineup synergy with Tellez and Garcia behind, and Langford before

Optimal Lineup vs. Right Hand Pitcher

#	Player	Pos.	OBP	wRC+	Obj
1	Mitch Garver	DH	.370	142	.0463
2	Corey Seager	SS	.372	152	.0486
3	Rowdy Tellez	1B	.315	115	.0401
4	Adolis Garcia	RF	.295	102	.360
5	Marcus Semien	2B	.322	108	.0356
6	Josh Jung	3B	.303	103	.0332
7	Leody Taveras	CF	.298	91	.0312
8	Jonah Heim	C	.286	83	.0283
9	Wyatt Langford	LF	.335	115	.0309

Obj. Value (IP)

Z = 0.4594

Solve Time: 12.8 seconds | LP Gap: 4.94%

Interpretation

Mitch Garver hits leadoff as his 142 WRC+ is the second highest only to Seager
Seager moves up significantly vs Righties

- Hits much better against RHP compared to LHP
 - Rowdy Tellez Plays 1B over Nate Lowe in both lineups
 - Both players are left-handed and have similar splits vs LHP and vs RHP
 - Tellez has more power with a higher SLG% and Iso despite slightly worse overall hitting stats
 - Model values the higher power hitter in Tellez to have better synergy with the rest of the lineup
- More "Traditional" compared to vs LHP

Key Takeaways

01 Main Optimization Result

The model produced two genuinely different lineups – 3 players were swapped for others in the scenarios, and Corey Seager dropped from the 2 spot to the 8.

Right-Handed Pitcher Lineup:

Garver → Seager → Tellez → Garcia → Semien → Jung → Taveras → Heim → Langford

Objective: 0.4594

Solve time: 12.8 seconds

Left-Handed Pitcher Lineup:

Garcia → Kelly → Pederson → Jung → Semien → Taveras → Langford → Seager → Tellez

Objective: 0.4629

Solve time: 8.1 seconds

02 Structural Insight

- The Model took 10,402 LP iterations, driven by synergy linearization and adding 225 $z_{i,j,s}$ variables which required 3 constraints each
- 9.2% of the integer columns went inactive when solving causing a model restart, reducing the model from 446 to 391 columns, This allowed to solver to identify these variables and remove them.
- Both the LHP and RHP solves were terminated at 1 node each, and required thousands of different heuristic and separation iterations, this means that although the branching tree was tight, the relaxation rates for the LP were quite weak

Key Takeaways Part 2

3 Sensitivity

- Gap of 4.94% (RHP) and 4.87% (LHP) – solutions are optimal within the 5% tolerance test
- Top of the order spots were more sensitive to pitcher handedness since the two lineups were completely different in that sense, bottom of the order spots were also volatile, as Seager dropped all the way from 2 to 8 all because of the platoon penalty we enforce.

4 Robustness Finding

- Spots 4-5 are the most stable – Jung and Semien remain close to these spots through both lineups, showing a balance in the splits
- 7 of 9 defensive positions stayed filled by the same position group, so defense remains robust despite roster change
- The model produced a feasible, near optimal solution for both scenarios.

5 What we learned

- Platoon Optimization is primarily a roster selection problem, whereas the big decisions were to include rather than where to bat them at.
- Split Shrinkage was essential – split wOBA along with overall wOBA gave us useful insight.
- The linking constraint was vital, co-optimizing batting order and defense together gave us results no other sequential approach could find.

6 Future Extensions

- Model matchups against specific pitchers, and not just the hand they pitch with.
- Extend to a full 162 game schedule (plus playoffs) to estimate cumulative wins added over a default lineup.
- Incorporate rest, fatigue, and injury availability to make the optimizer usable as a real daily lineup tool.